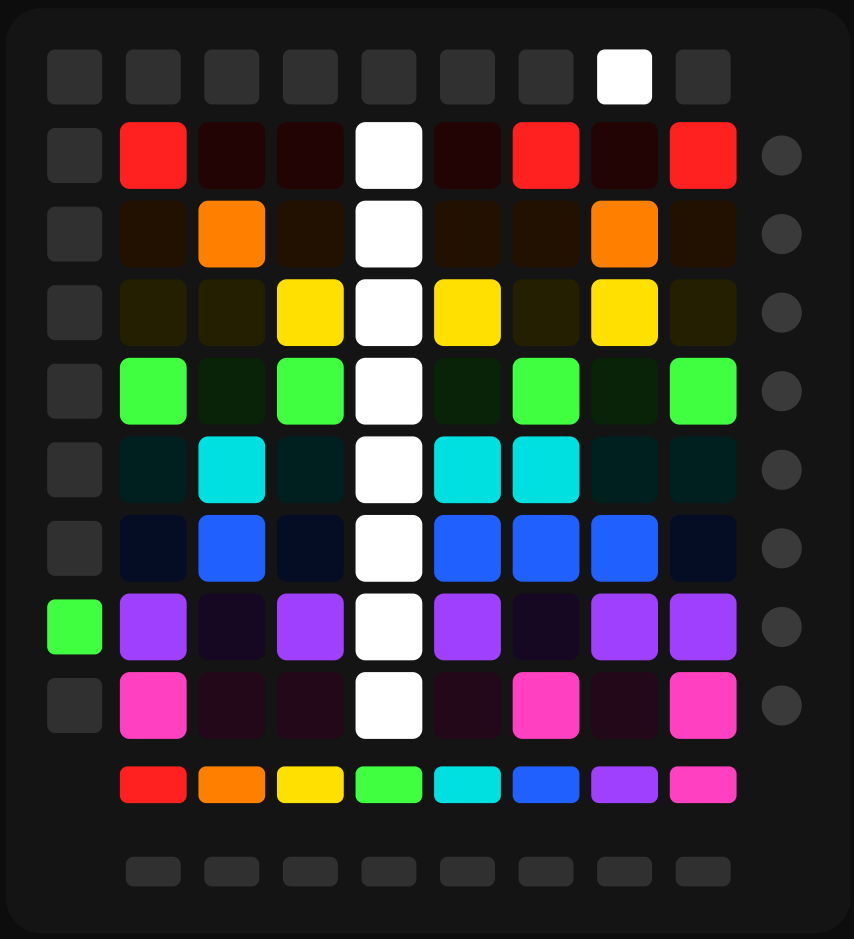
USER GUIDE

8-Track Sequencer

Standalone firmware for Launchpad Pro MK3



Early beta - flash at your own risk. Community firmware based on CoreFW (GPL-3.0). Not made, endorsed or supported by Novation or Focusrite. Launchpad and Launchpad Pro are trademarks of Focusrite Audio Engineering Ltd, used here only to identify the hardware.

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SECTION 1

Introduction

The 8-Track Sequencer turns a Launchpad Pro MK3 into a standalone MIDI sequencer. It runs on the Launchpad itself – no computer is needed once it is installed. Connect synths and drum machines to the two DIN MIDI outputs (or to a computer over USB) and program up to eight parts at once.

Key features

- **8 tracks**, each with its own MIDI channel, output (USB, DIN 1, DIN 2 or both DIN outs), note range, transpose and step rate (1/32 to a whole bar, including triplets)
- **16 patterns** per project, each holding all 8 tracks, of up to **64 steps** per track; switched and chained as a song
- **Chords** – up to 8 notes on every step
- Per-step **velocity**, **gate length** and **probability** (with a probability page)
- **Mutate**: shuffle a track's part, long press to get the original back
- Tempo 30–300 BPM with **tap tempo** and an on-grid **BPM readout**; **swing** from 50 % to 75 %
- **Live recording** from the pads or an external keyboard, with a one-bar count-in, quantised to the nearest step, and **step recording** one note at a time while stopped
- **MIDI thru**: a keyboard plays whichever track is selected
- Sends **MIDI clock**, or can follow an external clock from USB or DIN In
- **32 projects** saved in the Launchpad's memory
- Playback continues while you use the Launchpad's Note mode

What you need

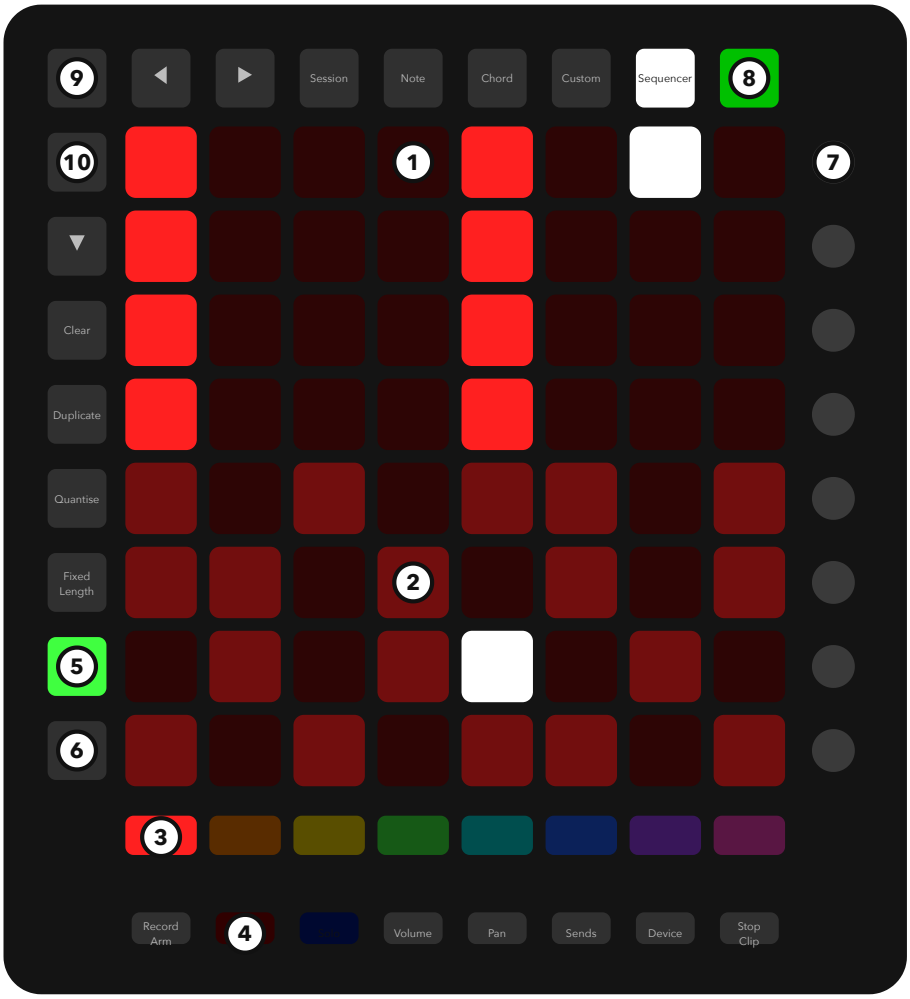
- A Launchpad Pro MK3 running the sequencer firmware (see *Installing the firmware* at the end of this guide)
- Synths, drum machines or a computer to play, connected to **MIDI Out**, **MIDI Out 2** or USB
- Optionally, a MIDI keyboard on **MIDI In** or the USB MIDI port

> **NOTE**
> The second DIN socket on the back of the Launchpad is used as **MIDI Out 2** by this firmware (not as MIDI Thru).

SECTION 2

Overview

The sequencer uses the whole surface of the Launchpad. The top half of the grid shows the steps of the selected track; the bottom half is a keyboard of 32 notes. The coloured row under the grid selects tracks and the row beneath that holds Mute and Solo.



The sequencer with track 1 selected and playing

1. **Step area.** 32 steps of the selected track's current pattern: steps 1–32, or 33–64 after ▶ (top row = steps 1–8 or 33–40). Lit steps play; brighter = louder. The white pad is the playhead.
2. **Note area.** 32 chromatic notes, lowest at the bottom left. Tap a note to hear it and to choose the note new steps get.
3. **Track buttons.** Tap to select a track, hold for its settings. Each track has its own colour.
4. **Mute and Solo.** Arm a mode in which the track buttons mute or solo tracks (see *Mute and Solo*). While a step is held, this whole row sets its gate length.
5. **Play.** Start / stop. Green = internal clock, orange = following an external clock.
6. **Record.** Arm recording: live while playing, step by step while stopped.
7. **Patterns.** Tap to open or close the pattern view (16 patterns on the top two rows).
8. **Projects.** Tap to save, hold to switch project. Green = saved, amber = unsaved changes.

9 **Shift.** Hold for tempo, swing, tap tempo and the clock setting.

10 **▲ / ▼.** Move the note area up or down an octave.

Entering and leaving the sequencer

Press Sequencer to open the sequencer. The first time after power-up, the last project you used is loaded automatically.

Press Note to return to the Launchpad's Note mode. **Playback keeps going** while you play in Note mode – the Sequencer button lights green to show it. Press Sequencer again to come back.

SECTION 3

Getting Started

This walkthrough programs a simple pattern and plays it on a synth connected to **MIDI Out**.

1. Connect your synth's MIDI In to the Launchpad's **MIDI Out** socket, and set the synth to receive on **MIDI channel 1**.
2. Press Sequencer. Track 1 (red) is selected. New tracks send on **MIDI Out**, with the MIDI channel equal to the track number.
3. Tap a pad in the note area. You hear the note on your synth – it is now the note new steps will use.
4. Tap steps 1, 5, 9 and 13 in the step area to turn them on.
5. Press ▷ (Play). The playhead moves across the steps and the synth plays.
6. Hold step 5 and tap two more note pads: step 5 is now a three-note chord.
7. Tap track button 2 (orange), choose a note and add some steps. Track 2 plays on MIDI channel 2.
8. Tap Projects to save. The button flashes white, then turns green.

> **TIP**
> Hold **Shift** at any time to see the current tempo on the grid. Use **Shift + ▲ / ▼** to change it, or tap **Sends** in time while holding Shift.

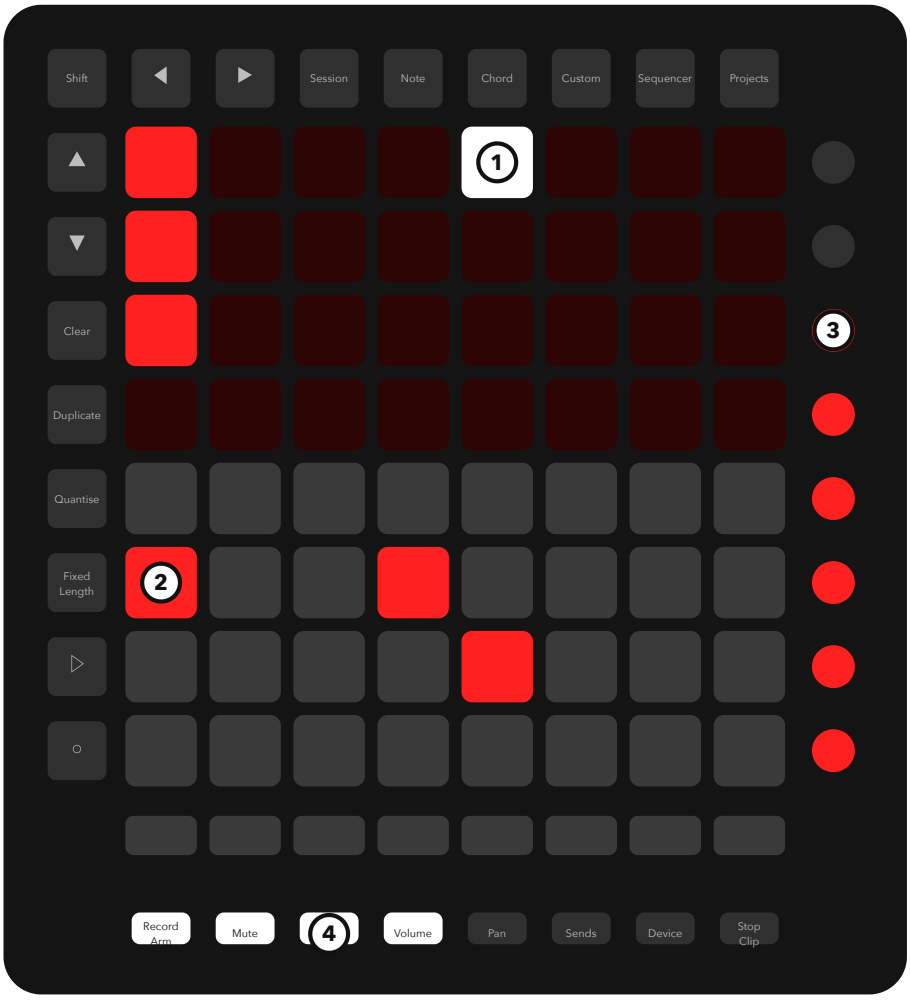
SECTION 4

Programming Steps

Action	Result
Tap a note pad	Choose the note new steps get, and hear it
Tap an unlit step	Turn it on with the chosen note
Tap a lit step and release	Turn it off
▲ / ▼	Move the note area up / down an octave

Holding a step

Hold a step to edit it. While it is held, the note area shows all its notes, the right-hand column shows its velocity and the bottom row shows its gate length.



Step 5 held: a three-note chord, velocity 96, gate 4/8

(1) **Held step.** Shown white while you hold it.

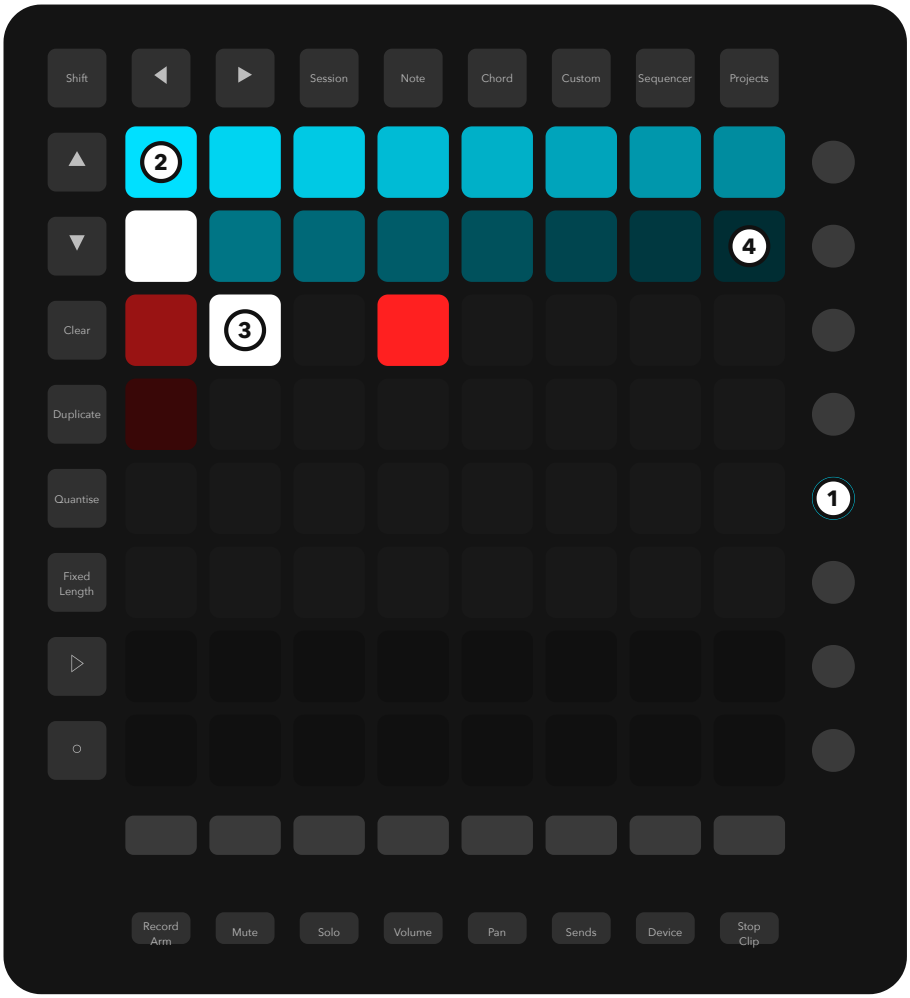
2. **Notes.** The first note pad you tap sets the step's note. Tap more pads in the same hold to build a chord of up to 8 notes; tap a lit note again to remove it.
3. **Velocity.** The right-hand column: top = 127, then 112, 96, 80, 64, 48, 32, bottom = 16.
4. **Gate length.** The bottom row: 1 (Record Arm) = shortest, 8 (Stop Clip) = the whole step.

Probability: hold the step and Shift – the right-hand column then sets how often the step plays: top = always (100 %), then 87, 75, 62, 50, 37, 25, bottom = 12 %.

> **TIP**
> A step you have edited while holding it never turns off when you let go.

Probability page

Tap **Probability** (right-hand column, 5th button down) to set how often the selected track's steps play: 16 levels on the top two rows, the steps below them.



Probability page: step 2 selected, set to level 9 (50 %)

1. **Probability.** Tap to open the page (cyan); tap again to go back to the steps.
2. **Levels.** The top two rows: level 1 (top-left) = every time (100 %) … level 16 = hardly ever (6 %). Brighter = more likely. The selected step's level is white.
3. **Steps.** Rows 3–6: steps 1–32 (◀ / ▶ for 33–64), dimmer the less likely they are. Tap a lit step to select it (white), then tap a level.
4. **Level 16.** Hardly ever (6 %). Off steps are faint and can't be selected; the playing step is in full colour.

Level	1	2	3	4	5	6	7	8
Plays	100 %	94 %	88 %	81 %	75 %	69 %	62 %	56 %

Level	9	10	11	12	13	14	15	16
Plays	50 %	44 %	38 %	31 %	25 %	19 %	12 %	6 %

Mutate

Mutation (right-hand column, 6th button down) rearranges the **selected track's** part of the current pattern for instant variations.

Action	Result
Tap Mutation	Shuffle: about a quarter of the lit steps (at least one) move to another step or swap notes with another lit step. Same hits and notes, new arrangement. The button turns purple.
Tap again	Shuffle further
Hold Mutation (about half a second)	**Back to the original**: the part as it was before the first tap. The button flashes white.

> **TIP**
> The original is remembered per track until you restore it, switch project or turn the Launchpad off. Mutating a different pattern on the same track remembers that one instead. Edits made while a part is mutated are lost when you restore; saving stores the part as it currently is.

Pages: steps 33–64

A pattern can have up to 64 steps; the step area shows 32 at a time. ◀ shows steps 1–32, ▶ shows steps 33–64. The arrow of the page you are on is white; the other arrow is grey when the pattern reaches that page, dim when it does not, and **green while the playhead is on that page**. To make a pattern longer than 32 steps, go to page 2, hold Fixed Length and tap the last step. Projects saved before 64 steps load unchanged.

Copying a step

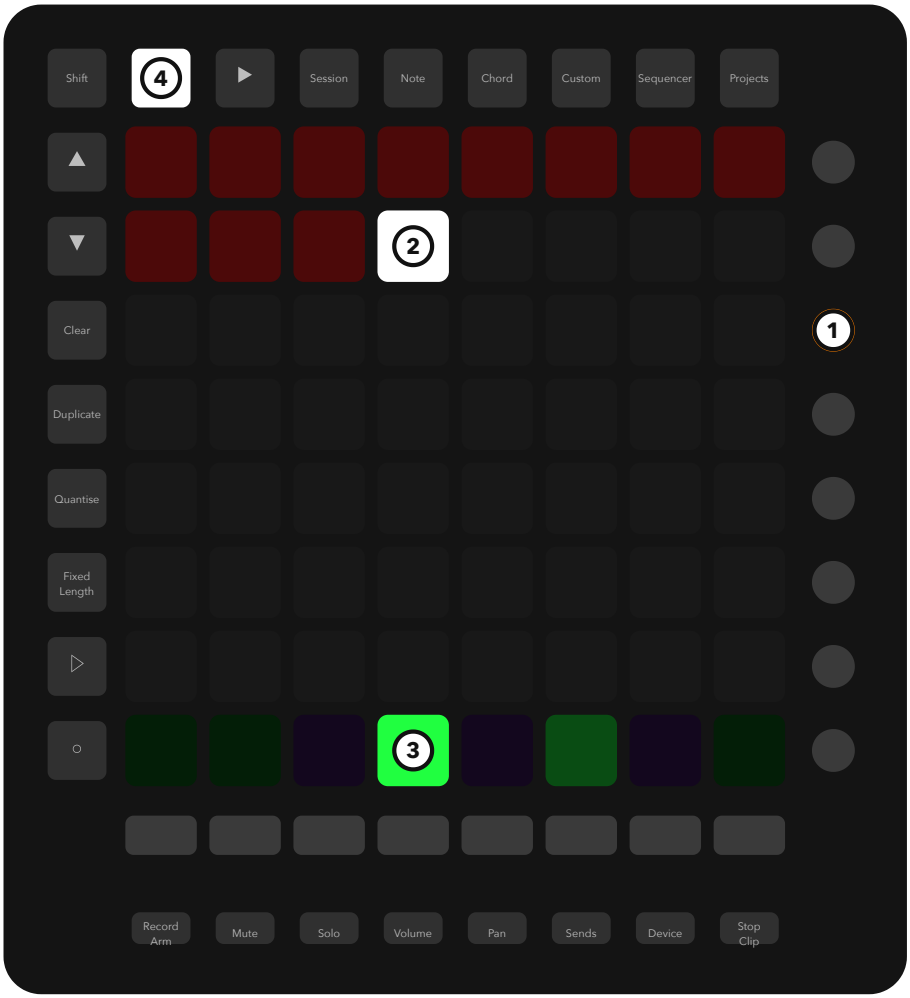
Hold Duplicate and tap a lit step: it lights **cyan**. Keep holding Duplicate and tap every step you want a copy on. Each copy gets the same notes (including chords), length, velocity and probability. Press ▶ / ◀ while still holding Duplicate to paste on the other page.

Pattern length and clearing

Action	Result
Hold Fixed Length + tap a step	The pattern ends at that step (1–32, or 33–64 on page 2)
Hold Clear + tap a step	Clear that step
Hold Clear + tap a track button	Clear that track's current pattern (its length stays)

Pattern Settings: length and step rate

Tap **Pattern Settings** (right-hand column, 3rd button down) to set the length and step rate of the selected track's current pattern. Every pattern has its own, so pattern 1 can be 8 steps at 1/8 and pattern 2 sixteen at 1/16.



Pattern Settings: 12 steps at 1/8

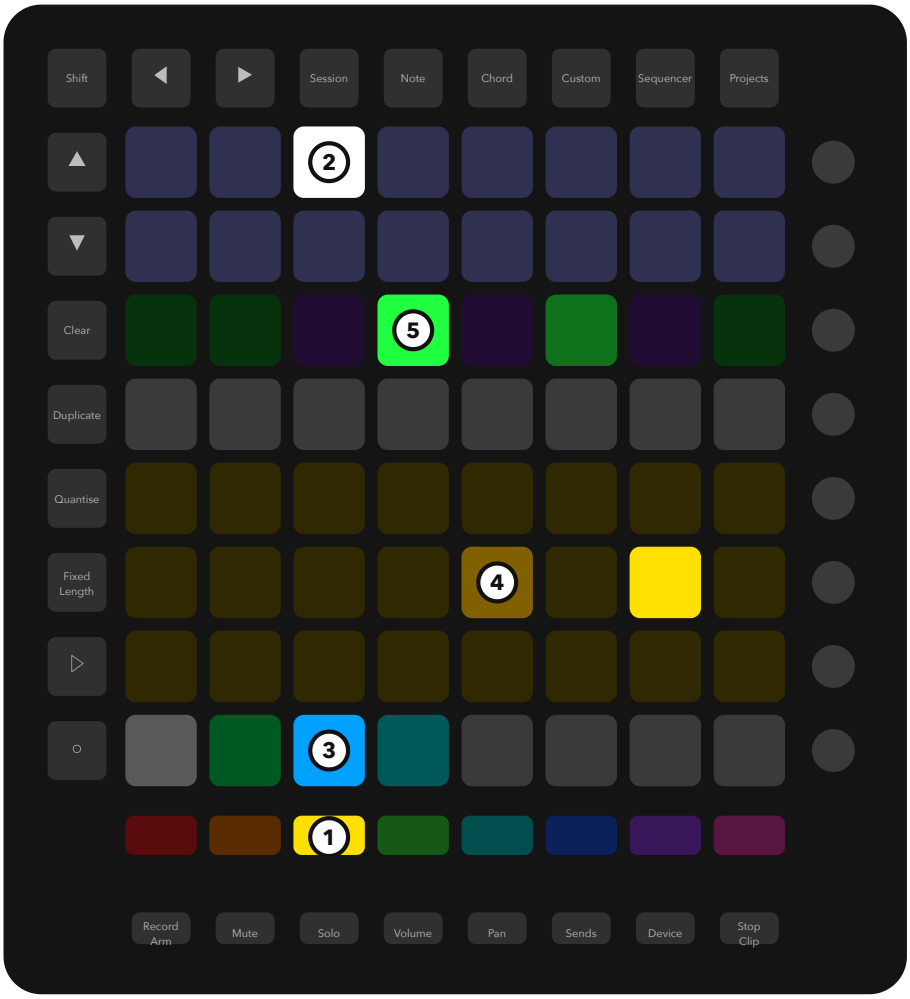
1. **Pattern Settings.** Tap to open the page (orange); tap again to go back to the steps.
2. **Length.** Tap a step pad to make it the last step (white). Steps in the pattern are lit, steps past its end dim.
3. **Step rate.** Bottom row, slowest on the left: 1/1, 1/4, 1/4T, 1/8, 1/8T, 1/16, 1/16T, 1/32. Straight green, triplets purple; the chosen one is bright.
4. **◀ / ▶.** Steps 1–32 / 33–64, for lengths over 32.

A new (empty) pattern starts with the length and rate of the pattern you were on. When a chain moves to a pattern at another rate, the new pattern starts exactly where the last step of the one before ends.

SECTION 5

Tracks

Tap a track button to select it. The step and note areas then show that track, and an external keyboard plays it (see *MIDI Thru*). Hold a track button to change the track's settings.



Track 3 held: MIDI channel 3, output MIDI Out 2, transposed +2, step rate 1/8

1. **Track button (held).** Settings are shown for as long as you hold it.
2. **MIDI channel.** Steps 1–16 (the top two rows) choose channel 1–16; the current channel is white. ▲ / ▼ also step the channel up and down.
3. **Output.** Bottom row of the note area, pads 1–4: USB (white), MIDI Out (green), MIDI Out 2 (blue), both DIN outs (cyan).
4. **Transpose.** Note rows 2–4: −12 … +11 semitones. The dim pad (row 3, pad 5) is 0.
5. **Step rate.** Steps 17–24 (the third row): how fast the track moves through its steps, slowest on the left, for all its patterns. Straight rates are green, triplets purple; the chosen rate is bright.

New tracks send on **MIDI Out** with MIDI channel = track number (track 3 = channel 3).

Step rate

Each track can step at its own speed against the shared tempo, for example a bass line at half speed under a 16th-note hi-hat. Hold the track button and tap a pad in the third row; the slowest rate is on the left. This sets **all** the track's patterns; to give one pattern its own rate, use *Pattern Settings*.

Pad	17	18	19	20	21	22	23	24
One step =	1 bar (1/1)	1/4	1/4 triplet	1/8	1/8 triplet	1/16 (default)	1/16 triplet	1/32
Speed	1/16×	¼×	⅜×	½×	¾×	1×	1.5×	2×

All tracks start together when you press Play, so tracks at different rates stay in line. Swing only affects tracks at 1/16; other rates play straight. Gate lengths scale with the step, and live recording uses the selected track's rate. Rates are saved with the project.

Mute and Solo

Mute and **Solo** in the bottom row arm a mode for the track buttons. The armed button flashes; press it again to disarm.

Press	Then tap a track button to
Mute (flashes red)	Mute or unmute it at once
Solo (flashes blue)	Solo or unsolo it at once
Shift + **Mute** (flashes red and white)	Mute or unmute it when its pattern next starts from step 1
Shift + **Solo** (flashes blue and white)	Solo or unsolo it when its pattern next starts from step 1

While Mute is armed, the track buttons show unmuted tracks in their colour and muted ones dark red; while Solo is armed, soloed tracks in their colour and the others dark. A queued change flashes the track button white until it happens; tap the track again to cancel it. When stopped, a queued change happens at once.

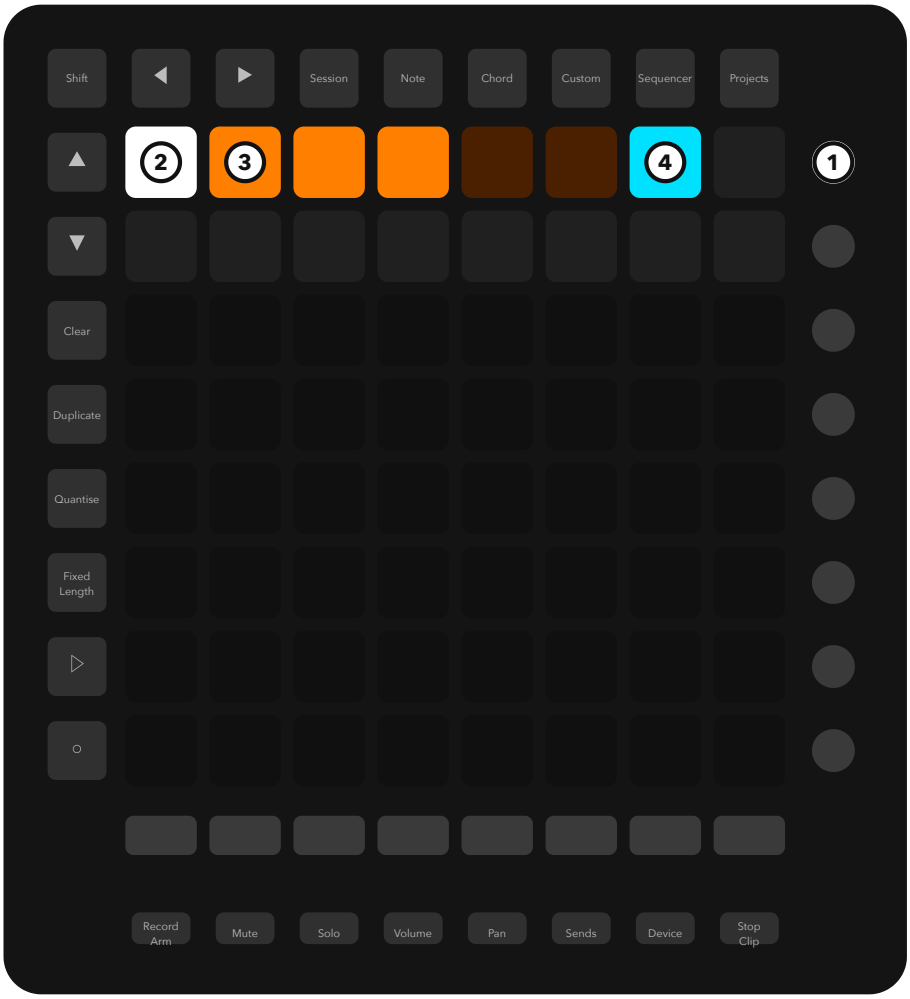
When any track is soloed, only soloed tracks play; a muted track stays silent even if soloed. Outside Mute and Solo mode, tracks silenced by mute or solo show darker. Mutes and solos are saved with the project.

SECTION 6

Patterns and Chains

A project has **16 patterns**. A pattern holds **all 8 tracks**, and every track is always on the same pattern: switching pattern switches the whole song.

Tap **Patterns** to open the pattern view: the button lights white and the top two grid rows show the 16 patterns (1–8 on the top row, 9–16 below). The view **stays open** until you tap **Patterns** again, so both hands are free to pick and chain patterns. You can also hold **Patterns**, tap a pattern and let go: the view then closes by itself.



Pattern view: pattern 1 playing, chain 1–4, pattern 7 queued

1. **Patterns.** Tap to open / close the view (white while open).
2. **Current pattern.** White.
3. **Chained.** Orange: part of the chain. Dim orange = has notes; dark = empty.
4. **Queued.** Cyan: starts when the playing pattern ends.

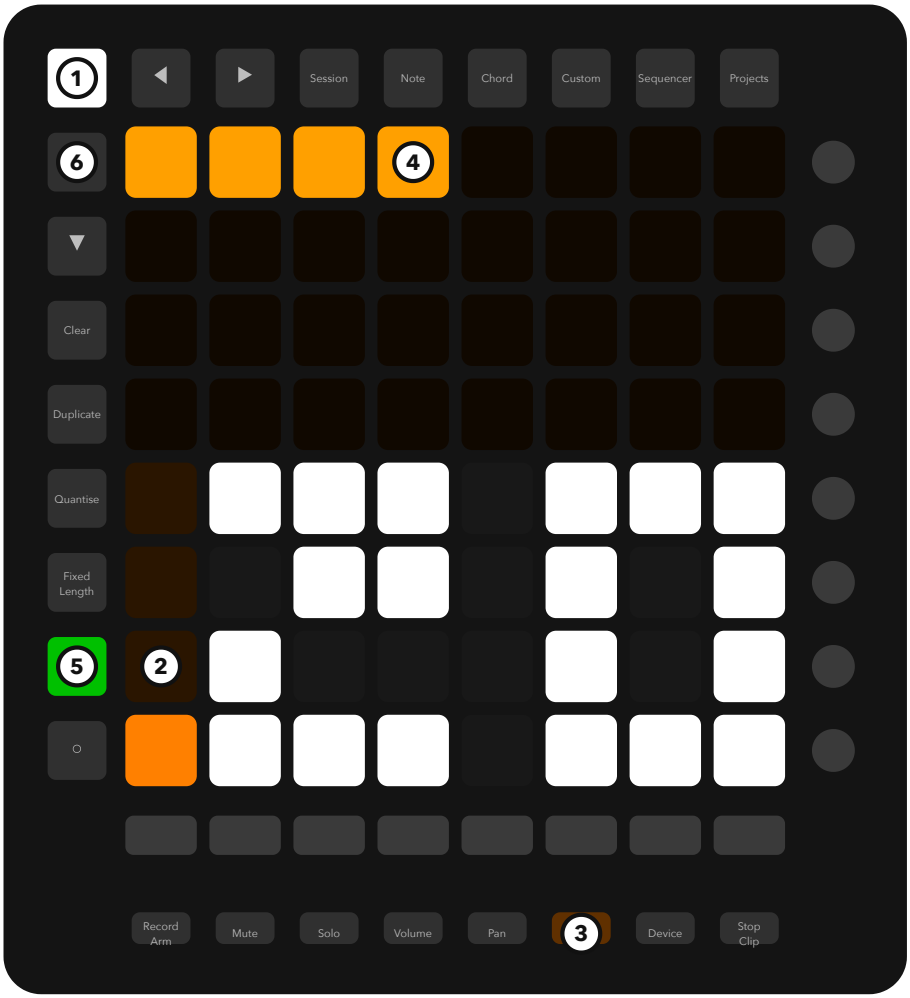
In the pattern view	Result
Tap a pattern	Switch to it – at the end of the playing pattern, or at once when stopped
Hold a pattern + tap another	**Chain** the range: 1–4 plays 1, 2, 3, 4, 1, …
Hold **Duplicate** + tap a pattern	Copy the current pattern (all tracks) there
Hold **Clear** + tap a pattern	Clear that pattern on all tracks (its length stays)

TIP

Each track's part of a pattern keeps its own length, so tracks can still run polymetric lengths. A new, empty pattern takes the lengths of the current one.

SECTION 7

Tempo, Swing and Playback



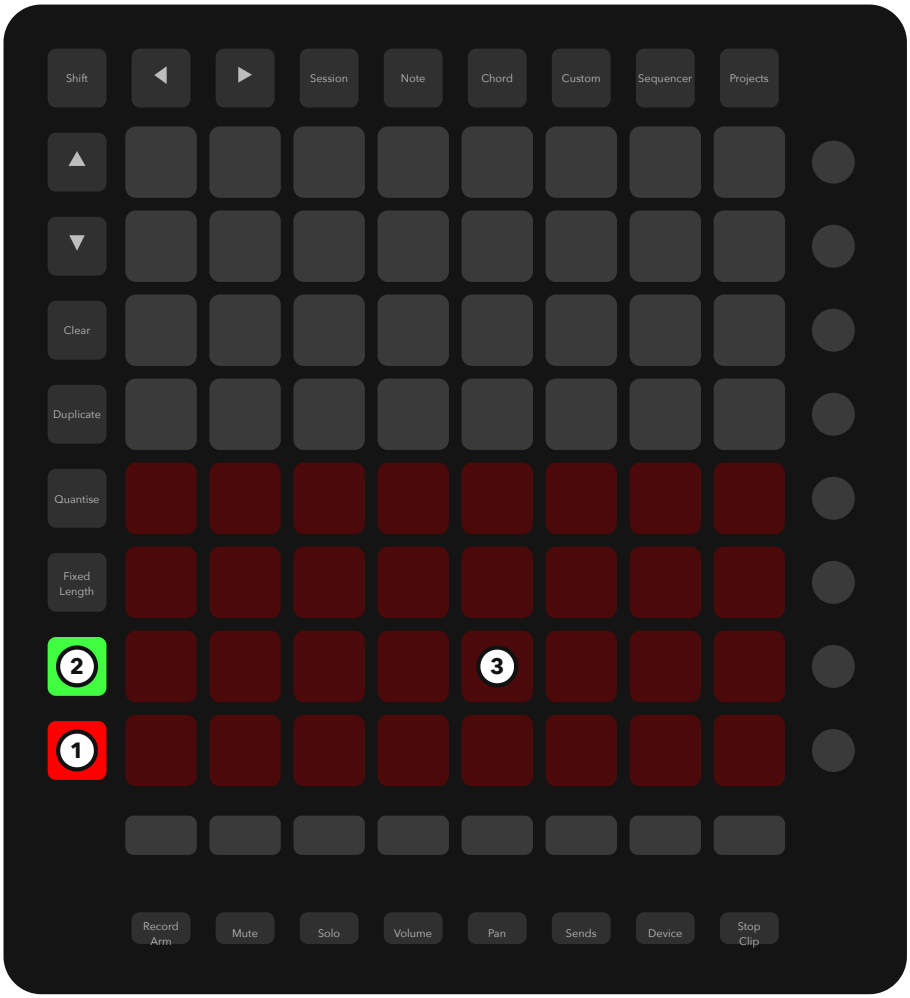
Shift held: 120 BPM on the note area, swing 53 %, Play green (internal clock)

1. **Shift.** Hold to see tempo, swing and the clock setting.
2. **BPM readout.** An orange bar for the hundreds (1–3 pads), then the tens and units in white.
3. **Tap.** Shift + Sends (printed *Tap*) is tap tempo.
4. **Swing.** The step area shows swing: step 1 = 50 % (straight) … step 26 = 75 %.
5. **Play.** With Shift: the external clock setting (see *MIDI Clock*).
6. **▲ / ▼.** With Shift: tempo.

Action	Result
▷	Start / stop
Shift + ▲ / ▼	Tempo +1 / −1 BPM (30–300)
Shift + **Fixed Length** + ▲ / ▼	Tempo +10 / −10 BPM
Shift + **Sends** (Tap)	Tap tempo: tap in time; the average of the last few taps sets the BPM. A 2-second pause starts over. The button flashes white on each tap.
Shift + a step 1–26	Swing 50–75 %

SECTION 8

Live Recording



Recording armed and playing

1. **Record.** Tap ○ or Record Arm (first button of the bottom row) to arm (dark red); bright red while playing.
2. **Play.** Recording happens while the sequencer plays.
3. **Notes.** Play the note pads, or a keyboard on MIDI In or USB.

Recorded notes land on the **nearest step** of the selected track's current pattern, with their velocity and their held length as the gate: a note held over several steps plays for as long as you held it. Notes played together are recorded as a **chord** of up to 8 notes, keeping the loudest velocity.

A note played just ahead of a step is not replayed on that pass – you have already heard it.

Tap ○ again to stop recording.

Count-in

With recording armed (○ or Record Arm), pressing ▷ first counts **one bar in**: the whole grid flashes **red 4 times**, one flash per beat at the current tempo, and recording starts on the next downbeat. Notes played during the count-in are not recorded; press ▷ again to cancel it.

> **TIP**
> The count-in is on by default. To switch it off or on, open the **Setup** page and tap the first pad of the second row from the bottom (bright red = on, dim = off). There is no count-in while following an external clock.

Step recording (while stopped)

Arm ○ while the sequencer is **stopped** to enter notes one step at a time:

1. Tap ○ (dark red). A **white cursor** appears on step 1.
2. Play a note on the note pads or a keyboard. It goes on the cursor step, with its velocity, and the cursor moves to the next step when you let go.
3. Keep playing: each note fills the next step. For a **chord**, hold the notes together; the cursor moves on once you have released them all.
4. To leave a **rest** (or go back and redo a step), tap the step you want next: the cursor jumps there, and the steps in between keep whatever they had.
5. After the pattern's **last step**, Record disarms by itself and the cursor returns to step 1.

Recorded steps get the default gate (half a step); set it afterwards by holding the step. The cursor follows onto page 2 for steps 33–64.

> **TIP**
> While Record is armed and the sequencer is stopped, tapping a step only moves the cursor. Disarm Record to switch steps on and off again. Press Play at any time to hear the result; Record stays armed, so it then records live.

SECTION 9

MIDI Thru

A keyboard connected to **MIDI In** or to the Launchpad's USB MIDI port plays the **selected track**. Notes, aftertouch, controllers and pitch bend are re-sent on that track's output and MIDI channel, whatever channel the keyboard sends on. Select another track and the keyboard plays that track's instrument.

Program changes and bank select are not passed on, so changing presets on the keyboard never changes the sound of the synth the selected track plays.

A note-off always goes to wherever its note-on went, so changing track while holding keys never leaves notes stuck.

> **TIP**
> If your keyboard has its own sounds, turn its **Local** setting off so you only hear the selected track's instrument.

If the keyboard has MIDI Thru switched on, the sequencer's own notes and controllers come back to it. These echoes are recognised and dropped, so they do not loop or record twice.

SECTION 10

MIDI Clock

Sending clock

When the sequencer runs on its own tempo, it sends MIDI clock, start and stop to USB, MIDI Out and MIDI Out 2, so drum machines and arpeggiators can follow it.

> **TIP**
> **Clock Out** can be switched off: open the **Setup** page and tap the second pad of the second row from the bottom, next to the count-in pad (bright green = on, the default; dim = off). With it off the sequencer sends no clock, start or stop, and an external clock it follows is not passed on either. It is remembered when you leave Setup.

Internal or external clock

By default the sequencer runs on its own clock and ignores any clock, start or stop arriving on **MIDI In** or the USB MIDI port; ▷ is **green**. Press Shift + ▷ to switch to following an external clock and back. While Shift is held, ▷ shows the setting:

▷ with Shift held	Setting
Green	Internal clock (default): ignore incoming clock, start and stop; run on and send the sequencer's own clock
Orange	Follow an external clock

Following an external clock

- ▷ flashes **orange**.
- The clock is passed on to both DIN outputs.
- Tempo controls and tap tempo have no effect.
- Half a second after the last clock pulse, the sequencer carries on with its own clock.

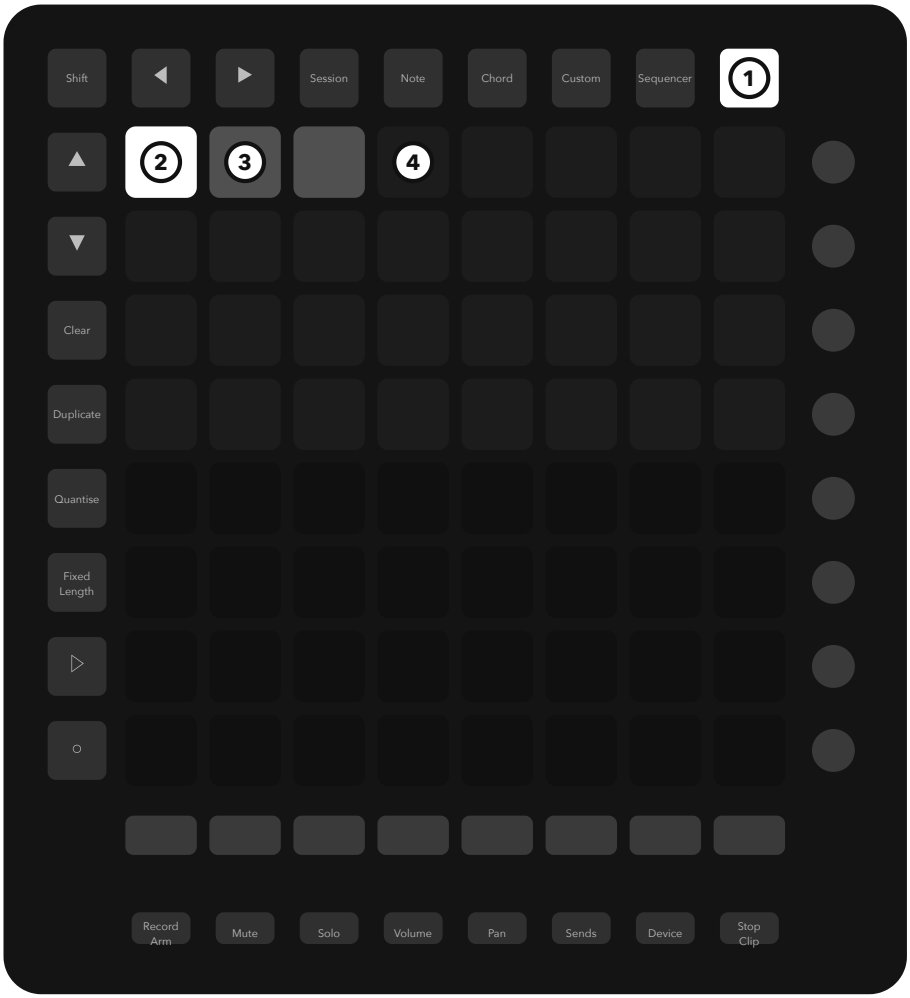
The setting is saved with the project; projects saved before it existed load on the internal clock.

> **TIP**
> Some synths (a Korg minilogue, for example) send a clock that stutters while they load presets. Only follow an external clock when you need to.

> **NOTE**
> A synth with MIDI Thru switched on sends the sequencer's clock straight back. The sequencer recognises its own clock and ignores it, so the tempo cannot run away.

SECTION 11

Projects



Projects held: project 1 current, projects 2 and 3 saved

1. **Projects.** Tap to save the current project (flashes white). Hold to see all 32.
2. **Current project.** White.
3. **Saved project.** Dim. Tap it (while holding Projects) to switch to it.
4. **Empty project.** Dark. Tap it to start a new, blank project there.

Switching saves the current project first. The Projects button is **green** when everything is saved and **amber** when there are unsaved changes.

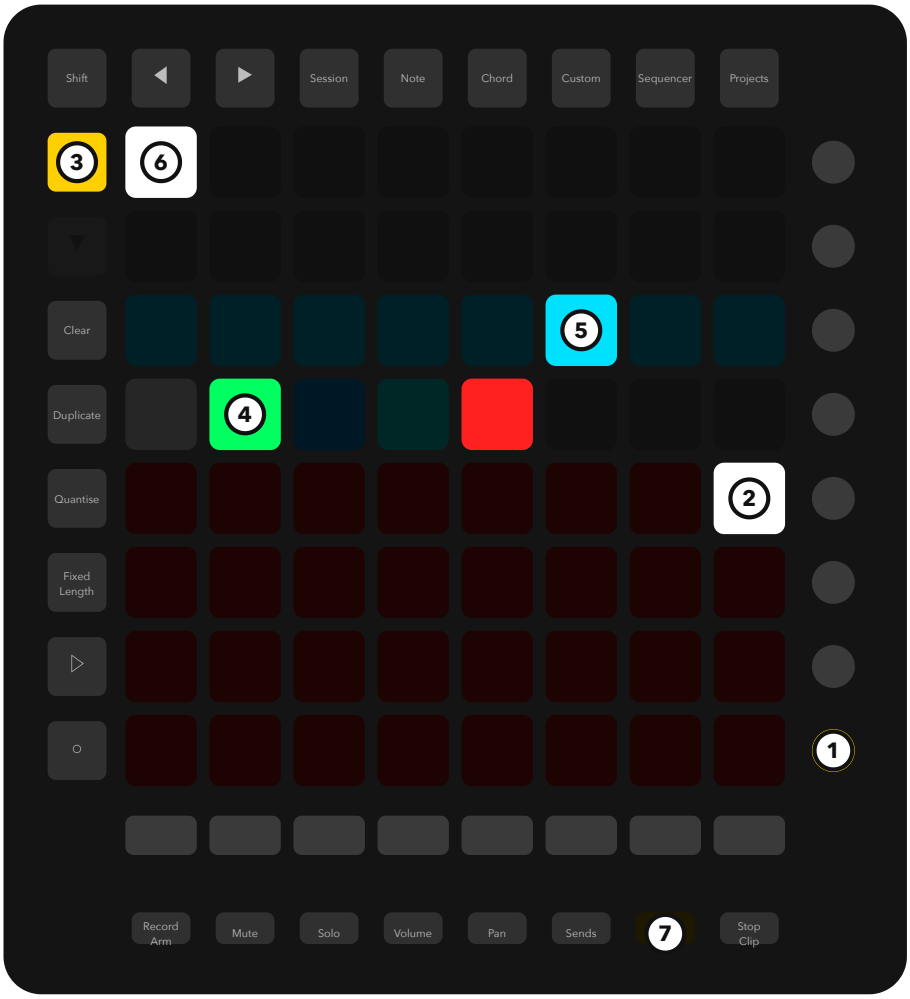
A project holds everything: all tracks, the 16 patterns, chains, channels, outputs, transpose, step rates, program changes, tempo, swing and the external clock setting. The current project opens automatically after power-up.

Only the steps in use are stored. Each project has room for about 2,500 used steps; a project fuller than that cannot be saved.

SECTION 12

Program Changes

A project can set up your synths when you select it: it sends a program change (and, if you want, a bank select) for each track, plus one more for the whole project. Tap **Print to Clip** (the bottom button of the right-hand column) to open the **Programs page**; it lights yellow. Tap it again to go back to the steps.



Programs page, track 1: bank 5, program 8, on MIDI Out, channel 1

1. **Print to Clip.** Opens and closes the Programs page.
2. **Programs.** The bottom four rows are 32 programs, read like text: program 1 top left, counting along each row and then down (25–32 on the bottom row). The chosen one is white. Tap it again for none (nothing is sent).
3. **▲ / ▼.** The next / previous 32 programs (1–32, 33–64, 65–96, 97–128); lit while there are more that way. Row 5, pads 5–8 show which 32 (and select them too).
4. **Output.** Row 5, pads 1–4: USB, MIDI Out, MIDI Out 2, both.
5. **Bank.** Row 6: bank 0–7, sent as bank select (CC 32) just before the program change. Tap the lit one again for no bank.
6. **MIDI channel.** The top two rows: channel 1–16.
7. **Device.** Bottom row, 7th button: edit the project slot instead of a track. Tap a track button to go back to the tracks.

The page edits the selected track. A track's program is sent on that track's output and channel (changing them here changes the track's). The **project slot** has its own output and channel. Use it for a message that is not tied to one track, such as choosing a performance on a multitimbral synth, whose global channel no track uses.

Choosing a program or bank sends it straight away, so you can try sounds. When you select a project in the Projects grid, it sends the project slot first, then each track that has a program; an identical message is only sent once. Tap the current project again in the grid to send them again (for example after switching a synth on). The programs are saved with the project; projects saved before this version send nothing.

Example: Novation Nova performance

Set the Nova's Global MIDI channel to one no Part uses (say 16) and its *Incoming program change* to Enabled. On the project slot choose that channel and bank 1 (Performance Bank A), then the performance number. Each Part can then get its own sound from a track on that Part's channel: bank 5 = Program Bank A, bank 6 = Program Bank B. Program 1 on the page is the Nova's number 000.

SECTION 13

Quick Reference

Control	Tap	With Shift	Hold + …
Step pad	Toggle step	Set swing (steps 1–26)	Note pads = notes / chord; right column = velocity (+ Shift = probability); bottom row = gate
Note pad	Choose and hear note	–	–
Track button	Select track	–	Steps 1–16 = channel; steps 17–24 = step rate; note row 1 = output; rows 2–4 = transpose; ▲ / ▼ = channel
Mute / Solo	Arm: track buttons mute / solo at once	Arm: changes wait for the pattern start	–
Sends	–	Tap tempo	–
▲ / ▼	Octave	Tempo ±1 (±10 with Fixed Length)	–
▷ **Play**	Start / stop (Record armed: 1-bar count-in)	Internal (green) / external clock (orange)	–
○ **Record / Record Arm**	Arm / disarm recording (stopped = step recording)	–	–
Fixed Length	–	–	+ step = pattern length
◀ / ▶	Steps 1–32 / 33–64	–	–
Clear	–	–	+ step = clear step; + track = clear pattern
Duplicate	–	–	+ step = copy it, then + steps = paste; with Patterns: copy pattern
Mutation	Shuffle the selected track's part	–	Long press: back to the original
Pattern Settings	Open / close the Pattern Settings page (length, step rate)	–	–
Probability	Open / close the probability page (16 levels; tap a step, then a level)	–	–
Print to Clip	Open / close the Programs page (program changes sent when a project is selected)	–	–

Control	Tap	With Shift	Hold + …
Patterns	Open / close the pattern view: switch, chain, copy, clear	–	Same; closes when released
Projects	Save	–	Top four rows = 32 projects: switch (the current one: re-send its programs)
Note	Note mode (playback continues)	–	–

SECTION 14

Installing the Firmware

The sequencer is part of a custom build of the open-source CoreFW firmware (github.com/anthonyhfm/launchpad-core-firmware, GPL-3.0). No Novation firmware is supplied with it.

> **WARNING**
> This firmware is an early beta, tested on very few devices. Flashing your own Launchpad is at your own risk: the firmware may misbehave or lose your projects and settings, and an interrupted or failed update can leave the Launchpad needing recovery from the bootloader. The authors accept no responsibility for damage to your device or data. It is not Novation software and Novation does not support it.

Installing

1. Download the latest `.syx` file from github.com/cyberheater/Launchpad-Pro-MK3-Custom-firmware/releases, or build it yourself: `cargo lppmk3` in that repository (branch `lppmk3-sequencer`) makes `build/lppmk3/fw.syx`.
2. Unplug the Launchpad. Hold **Setup** while plugging the USB cable back in. The Launchpad starts in bootloader mode and a MIDI port with "Bootloader" in its name appears.
3. Send the `.syx` file to that port. Use CoreFW's `python3 tools/flash.py <file>.syx` (needs `pip install mido python-rtmidi`), the Mac-only `swift scripts/flash_syx.swift <file>.syx` from github.com/cyberheater/lpp-emu, or any SysEx tool (SysEx Librarian, MIDI-OX). Do not unplug the Launchpad while it is sending.
4. The Launchpad restarts with the sequencer firmware. Press **Sequencer** to open it.

Returning to the original firmware

Enter bootloader mode the same way and update the Launchpad with Novation Components.

> **NOTE**
> Projects are stored separately from the firmware, so updating the sequencer firmware should keep them. Projects saved by earlier sequencer versions load in later ones.